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Supplementary Materials for

HMRF-AVAC: An Inverse Kinematics Solving and Force Control Method for Redundant Arms Based on Hybrid-Prior KAN

Zhuoqi Man, Junming Zhang, Junyu Wu, Yubin Liu,

Guoqing Chu, Jie Zhao, *Member IEEE*

This is supplementary material for the paper " **HMRF-AVAC: An Inverse Kinematics Solving and Force Control Method for Redundant Arms Based on Hybrid-Prior KAN** ", intended to supplement the relevant details mentioned in the paper.

The PDF file includes:

S1. Discussion on node information of the HMRF-KAN prior branch network structure in Section *II-C* (*HMRF-KAN*).

S2. Detailed discussion on the modified DH parameters of 9-DOF manipulator structure in Section *II-E* (*Network Training and Performance Evaluation*).

In order to facilitate the reader to find the content, we have listed the pages of the corresponding content here.

S1 ： page 2

S2 ： page 11

S1：Discussion on node information of the HMRF-KAN prior branch network structure in II-C

Fig. 1 shows the structure of HMRF-KAN. Fig.2-Fig.9 present the complete structures of joint prior blocks in prior branch network, Table II-Table IX list the node information associated with each prior block.

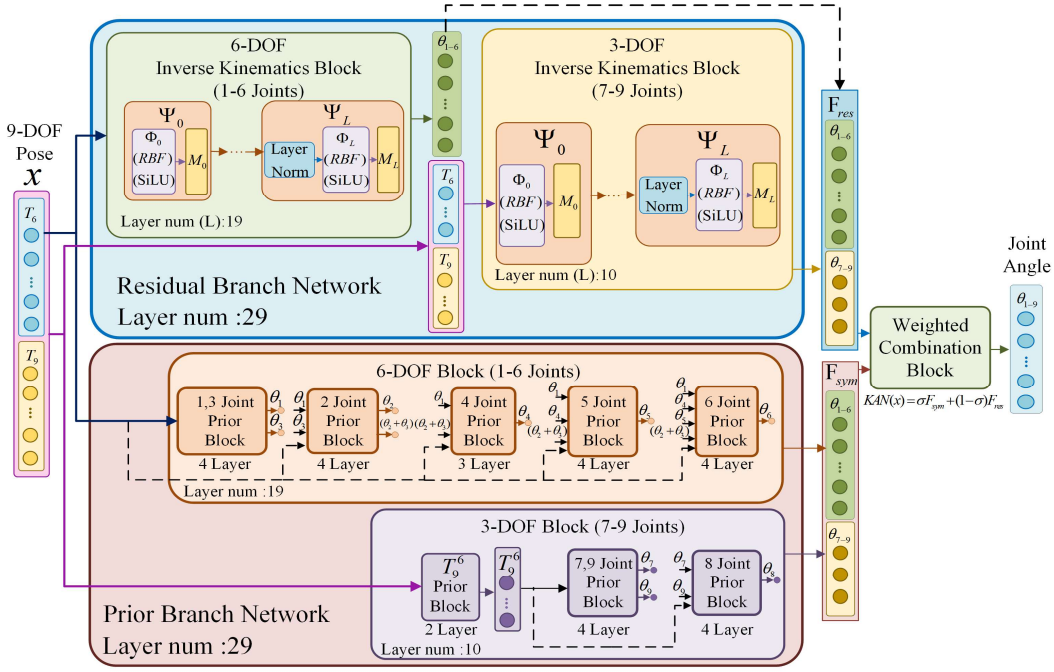


Fig. 1. The Network Structure of HMRF-KAN

The structure and node information of the 1,3 joint prior block:

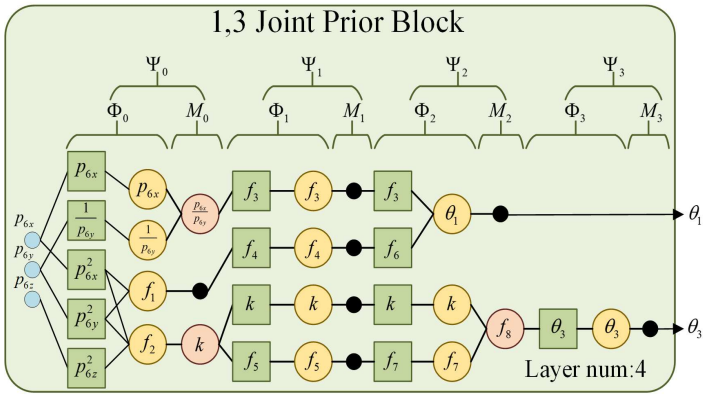


Fig. 2. The 1,3 Joint Prior Block

TABLE II

THE NODE INFORMATION OF 1,3 JOINT PRIOR BLOCK

Ψ_L: HMRF-KAN Layers
Φ_L: KAN Layers
M_L: Multiplication Layers

Output		θ_1			θ_3
Ψ_3	M_3	θ_1			θ_3
	Φ_3	θ_1			θ_3
		θ_1			$\theta_3 = \text{atan2}(a_3, d_4) - \text{atan2}(k, \sqrt{a_3^2 + d_4^2 - k^2})$
Ψ_2	M_2	θ_1			$\frac{k}{\sqrt{a_3^2 + d_4^2 - k^2}}$
	Φ_2	$\theta_1 = \text{atan2}(p_{6y}, p_{6x}) - \text{atan2}(d_2, \sqrt{p_{6x}^2 + p_{6y}^2 - d_2})$		k	$\frac{1}{\sqrt{a_3^2 + d_4^2 - k^2}}$
		$\text{atan2}(p_{6y}, p_{6x})$	$\text{atan2}(d_2, \sqrt{p_{6x}^2 + p_{6y}^2 - d_2^2})$	k	$\frac{1}{\sqrt{a_3^2 + d_4^2 - k^2}}$
Ψ_1	M_1	$\text{atan2}(p_{6y}, p_{6x})$	$\sqrt{p_{6x}^2 + p_{6y}^2 - d_2^2}$	k	$\sqrt{a_3^2 + d_4^2 - k^2}$
	Φ_1	$\text{atan2}(p_{6y}, p_{6x})$	$\sqrt{p_{6x}^2 + p_{6y}^2 - d_2^2}$	k	$\sqrt{a_3^2 + d_4^2 - k^2}$
		$\text{atan2}(p_{6y}, p_{6x})$	$\sqrt{p_{6x}^2 + p_{6y}^2 - d_2^2}$	k	$\sqrt{a_3^2 + d_4^2 - k^2}$
Ψ_0	M_0	$\frac{P_{6x}}{P_{6y}}$	$(P_{6x}^2 + P_{6y}^2 - d_2^2)$	$k = (\frac{p_{6x}^2 + p_{6y}^2 + p_{6z}^2 - a_2^2 - a_3^2 - d_2^2 - d_4^2}{2a_2})$	
	Φ_0	P_{6x} $(P_{6y})^{-1}$	$(P_{6x}^2 + P_{6y}^2 - d_2^2)$	$(p_{6x}^2 + p_{6y}^2 + p_{6z}^2 - a_2^2 - a_3^2 - d_2^2 - d_4^2)$	
		P_{6x} $(P_{6y})^{-1}$	P_{6x}^2	P_{6y}^2	P_{6z}^2
Input Parameters		P_{6x} P_{6y} P_{6z}			

The structure and node information of the 2 joint prior block:

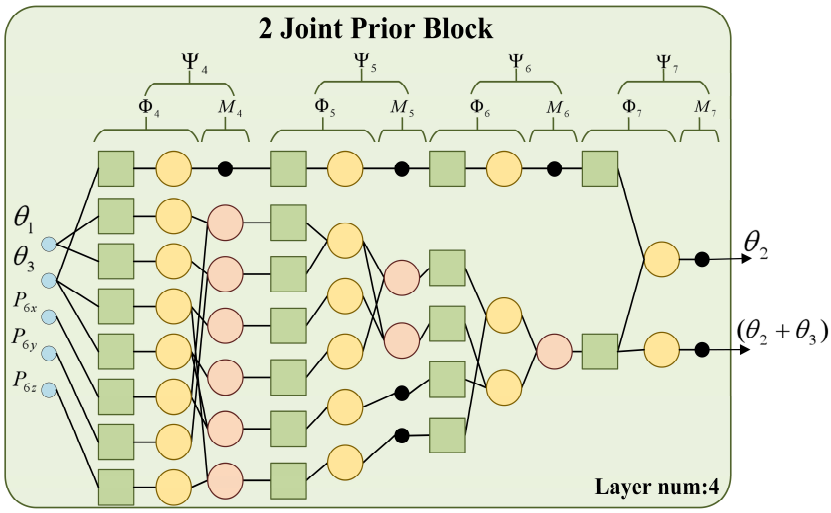


Fig. 3. The 2 Joint Prior Block

TABLE III

THE NODE INFORMATION OF 2 JOINT PRIOR BLOCK

Ψ_L: HMRF-KAN Layers
Φ_L: KAN Layers
M_L: Multiplication Layers

		Node Information						
Output		θ_2	$(\theta_2+\theta_3)$					
Ψ_7	M_7	θ_2	$(\theta_2+\theta_3)$					
	Φ_7	$\theta_2=(\theta_2+\theta_3)-\theta_3$	$(\theta_2+\theta_3)$					
		θ_3	$(\theta_2+\theta_3)=\text{atan2}(f_4,f_5^{-1})$					
Ψ_6	M_6	θ_3	$f_4\cdot f_5$					
	Φ_6	θ_3	$f_4=f_2-(a_2\cos(\theta_3)+a_3)P_{6z}$			$f_5=(f_3+(a_2\sin(\theta_3)-d_4)P_{6x})^{-1}$		
		θ_3	f_2		f_3	$(a_2\sin(\theta_3)-d_4)P_{6x}$	$(a_2\cos(\theta_3)+a_3)P_{6z}$	
Ψ_5	M_5	θ_3	$f_2=f_1\cdot(a_2\sin(\theta_3)-d_4)$		$f_3=f_1\cdot(a_2\cos(\theta_3)+a_3)$	$(a_2\sin(\theta_3)-d_4)P_{6x}$	$(a_2\cos(\theta_3)+a_3)P_{6z}$	
	Φ_5	θ_3	$f_1=\cos(\theta_1)P_{6x}+\sin(\theta_1)P_{6y}$	$a_2\cos(\theta_3)+a_3$	$a_2\sin(\theta_3)-d_4$	$(a_2\sin(\theta_3)-d_4)\cdot P_{6x}$	$(a_2\cos(\theta_3)+a_3)\cdot P_{6z}$	
		θ_3	$\cos(\theta_1)P_{6x}$	$\sin(\theta_1)P_{6y}$	$a_2\cos(\theta_3)+a_3$	$a_2\sin(\theta_3)-d_4$	$(a_2\sin(\theta_3)-d_4)\cdot P_{6x}$	$(a_2\cos(\theta_3)+a_3)\cdot P_{6z}$
Ψ_4	M_4	θ_3	$\cos(\theta_1)P_{6x}$	$\sin(\theta_1)P_{6y}$	$a_2\cos(\theta_3)+a_3$	$a_2\sin(\theta_3)-d_4$	$(a_2\sin(\theta_3)-d_4)\cdot P_{6x}$	$(a_2\cos(\theta_3)+a_3)\cdot P_{6z}$
	Φ_4	θ_3	$\cos(\theta_1)$	$\sin(\theta_1)$	$a_2\cos(\theta_3)+a_3$	$a_2\sin(\theta_3)-d_4$	P_{6x}	P_{6y} P_{6z}
		θ_3	$\cos(\theta_1)$	$\sin(\theta_1)$	$a_2\cos(\theta_3)$	$a_2\sin(\theta_3)$	P_{6x}	P_{6y} P_{6z}
Input Parameters		θ_1 θ_3 P_{6x} P_{6y} P_{6z}						

The structure and node information of the 4 joint prior block:

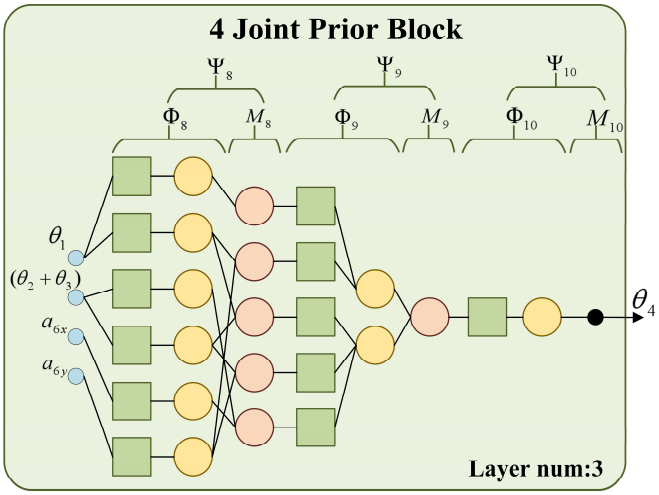


Fig. 4. The 4 Joint Prior Block

TABLE IV

THE NODE INFORMATION OF 4 JOINT PRIOR BLOCK

Ψ_L: HMRF-KAN Layers Φ_L: KAN Layers M_L: Multiplication Layers		Node Information					
Output		θ_4					
Ψ_{10}	M_{10}	θ_4					
	Φ_{10}	θ_4					
		$\theta_4 = \text{atan2}(f_1, f_5)$					
Ψ_9	M_9	$f_1 \cdot f_5^{-1}$					
	Φ_9	$f_1 = -a_4 s_1 + a_{6y} c_1$			$f_2 = -a_{6x} c_1 c_{23} - a_{6y} s_1 c_{23} + a_{6x} s_{23}$		
		$-a_4 s_1$	$a_{6y} c_1$	$-a_{6x} c_1 c_{23}$	$-a_{6y} s_1 c_{23}$	$a_{6x} s_{23}$	
Ψ_8	M_8	$-a_4 s_1$	$a_{6y} c_1$	$-a_{6x} c_1 c_{23}$	$-a_{6y} s_1 c_{23}$	$a_{6x} s_{23}$	
	Φ_8	s_1	c_1	s_{23}	c_{23}	a_{6x}	a_{6y}
		$s_1 = \sin(\theta_1)$	$c_1 = \cos(\theta_1)$	$s_{23} = \sin(\theta_2 + \theta_3)$	$c_{23} = \cos(\theta_2 + \theta_3)$	a_{6x}	a_{6y}
Input Parameters		θ_1	$(\theta_2+\theta_3)$	a_{6x}	a_{6y}		

The structure and node information of the 5 joint prior block:

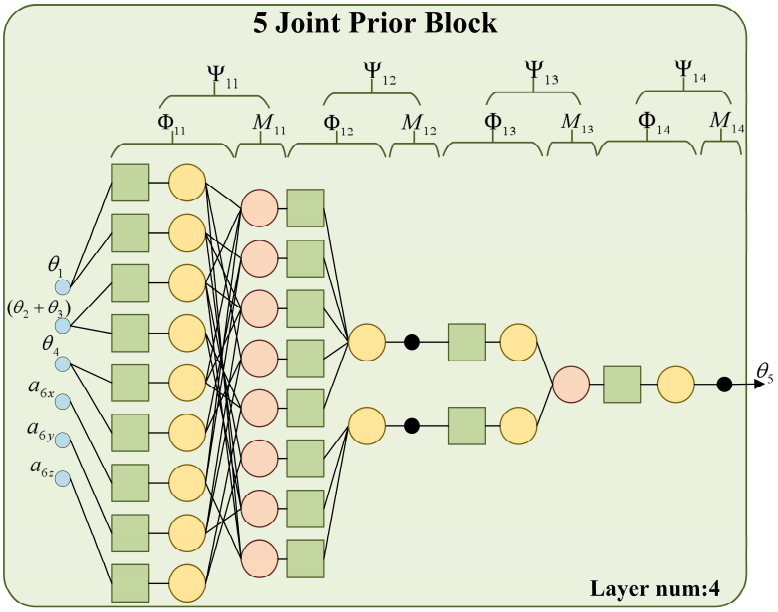


Fig. 5. The 5 Joint Prior Block

TABLE V

THE NODE INFORMATION OF 5 JOINT PRIOR BLOCK

Ψ_L: HMRF-KAN Layers Φ_L: KAN Layers M_L: Multiplication Layers		Node Information								
Output		θ_5								
Ψ_{14}	M_{14}	θ_5								
	Φ_{14}	θ_5								
		$\theta_5 = \text{atan}2(s_5, c_5)$								
Ψ_{13}	M_{13}	$s_5 \cdot c_5^{-1}$								
	Φ_{13}	s_5					c_5^{-1}			
		s_5					c_5^{-1}			
Ψ_{12}	M_{12}	s_5					c_5			
	Φ_{12}	$s_5 = a_{6x}c_1c_{23}c_4 + a_{6x}s_1s_4 + a_{6y}s_1c_{23}c_4 - a_{6y}c_1s_4 - a_{6z}s_{23}c_4$					$c_5 = a_{6x}c_1s_{23} + a_{6y}s_1s_{23} + a_{6z}c_{23}$			
		$a_{6x}c_1c_{23}c_4$	$a_{6x}s_1s_4$	$a_{6y}s_1c_{23}c_4$	$-a_{6y}c_1s_4$	$-a_{6z}s_{23}c_4$	$a_{6x}c_1s_{23}$	$a_{6y}s_1s_{23}$	$a_{6z}c_{23}$	
Ψ_{11}	M_{11}	$a_{6x}c_1c_{23}c_4$	$a_{6x}s_1s_4$	$a_{6y}s_1c_{23}c_4$	$-a_{6y}c_1s_4$	$-a_{6z}s_{23}c_4$	$a_{6x}c_1s_{23}$	$a_{6y}s_1s_{23}$	$a_{6z}c_{23}$	
	Φ_{11}	s_1	c_1	s_{23}	c_{23}	s_4	c_4	a_{6x}	a_{6y}	a_{6z}
		$s_1 = \sin(\theta_1)$	$c_1 = \cos(\theta_1)$	$s_{23} = \sin(\theta_2 + \theta_3)$	$c_{23} = \cos(\theta_2 + \theta_3)$	$s_4 = \sin(\theta_4)$	$c_4 = \cos(\theta_4)$	a_{6x}	a_{6y}	a_{6z}
Input Parameters		θ_1	$(\theta_2+\theta_3)$	θ_4	a_{6x}	a_{6y}	a_{6z}			

The structure and node information of the 6 joint prior block:

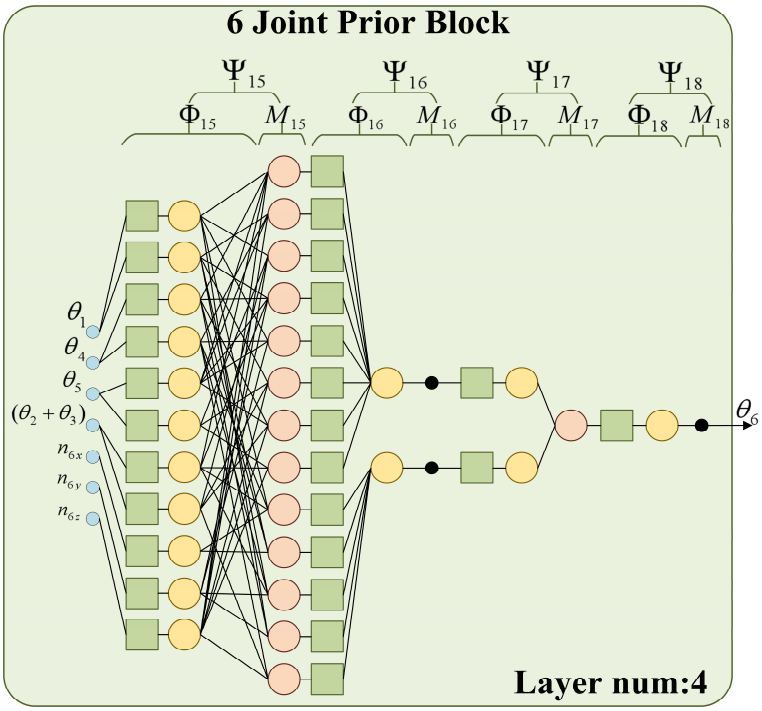


Fig. 6. The 6 Joint Prior Block

TABLE VI

THE NODE INFORMATION OF 6 JOINT PRIOR BLOCK

Ψ_L: HMRF-KAN Layers Φ_L: KAN Layers M_L: Multiplication Layers		Node Information
Output		θ_6
Ψ_{18}	M_{18}	θ_6
	Φ_{18}	θ_6
		$\theta_6 = \operatorname{atan2}(s_6, c_6)$
Ψ_{17}	M_{17}	$s_6 c_6^{-1}$
	Φ_{17}	c_6^{-1} s_6
		c_6^{-1} s_6
Ψ_{16}	M_{16}	c_6 s_6
	Φ_{16}	$c_6 = (n_{6x}c_1c_{23}c_4c_5 + n_{6x}s_1s_4c_5 - n_{6x}c_1s_{23}s_5 + n_{6y}s_1c_{23}c_4c_5 - n_{6y}c_1s_4c_5 - n_{6y}s_1s_{23}s_5 - n_{6z}s_{23}c_4c_5 - n_{6z}c_{23}s_5)$ $s_6 = -n_{6x}c_1c_{23}s_4 + n_{6x}s_1c_4 - n_{6y}s_1c_{23}s_4 - n_{6y}c_1c_4 + n_{6z}s_{23}s_4$
		$n_{6x}c_1c_{23}c_4c_5$ $n_{6x}s_1s_4c_5$ $-n_{6x}c_1s_{23}s_5$ $n_{6y}s_1c_{23}c_4c_5$ $-n_{6y}c_1s_4c_5$ $-n_{6y}s_1s_{23}s_5$ $-n_{6z}s_{23}c_4c_5$ $-n_{6z}c_{23}s_5$ $-n_{6x}c_1c_{23}s_4$ $n_{6x}s_1c_4$ $-n_{6y}s_1c_{23}s_4$ $-n_{6y}c_1c_4$ $n_{6z}s_{23}s_4$
Ψ_{15}	M_{15}	$n_{6x}c_1c_{23}c_4c_5$ $n_{6x}s_1s_4c_5$ $-n_{6x}c_1s_{23}s_5$ $n_{6y}s_1c_{23}c_4c_5$ $-n_{6y}c_1s_4c_5$ $-n_{6y}s_1s_{23}s_5$ $-n_{6z}s_{23}c_4c_5$ $-n_{6z}c_{23}s_5$ $-n_{6x}c_1c_{23}s_4$ $n_{6x}s_1c_4$ $-n_{6y}s_1c_{23}s_4$ $-n_{6y}c_1c_4$ $n_{6z}s_{23}s_4$
	Φ_{15}	c_1 s_1 c_4 s_4 c_5 s_5 c_{23} s_{23} n_{6x} n_{6y} n_{6z}
		$c_1 = \cos(\theta_1)$ $s_1 = \sin(\theta_1)$ $c_4 = \cos(\theta_4)$ $s_4 = \sin(\theta_4)$ $c_5 = \cos(\theta_5)$ $s_5 = \sin(\theta_5)$ $c_{23} = \cos(\theta_2 + \theta_3)$ $s_{23} = \sin(\theta_2 + \theta_3)$ n_{6x} n_{6y} n_{6z}
Input Parameters		θ_1 θ_4 θ_5 $(\theta_2+\theta_3)$ n_{6x} n_{6y} n_{6z}

The structure and node information of the T_9^6 prior block:

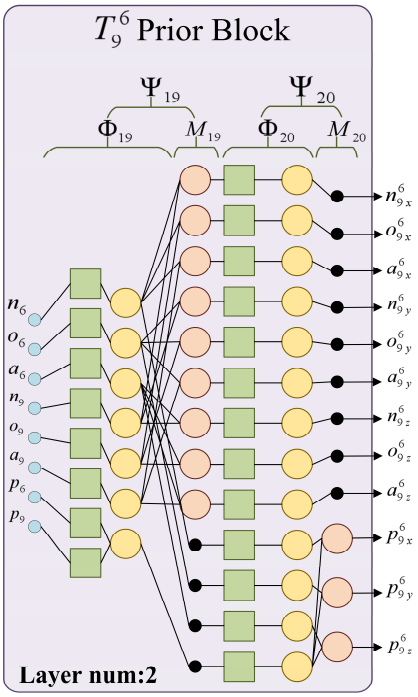


Fig. 7. The T_9^6 Prior Block

TABLE VII

THE NODE INFORMATION OF T_9^6 PRIOR BLOCK

Ψ_L: HMRF-KAN Layers Φ_L: KAN Layers M_L: Multiplication Layers		Node Information												
Output		n_{9x}^6	o_{9x}^6	a_{9x}^6	n_{9y}^6	o_{9y}^6	a_{9y}^6	n_{9z}^6	o_{9z}^6	a_{9z}^6	a_{9x}^6	p_{9x}^6	p_{9y}^6	p_{9z}^6
Ψ_{20}	M_{20}	n_{9x}^6	o_{9x}^6	a_{9x}^6	n_{9y}^6	o_{9y}^6	a_{9y}^6	n_{9z}^6	o_{9z}^6	a_{9z}^6	$p_{9x}^6 = n_6(p_9 - p_6)$		$p_{9y}^6 = o_6(p_9 - p_6)$	$p_{9z}^6 = a_6(p_9 - p_6)$
	Φ_{20}	n_{9x}^6	o_{9x}^6	a_{9x}^6	n_{9y}^6	o_{9y}^6	a_{9y}^6	n_{9z}^6	o_{9z}^6	a_{9z}^6	n_6	o_6	a_6	$(p_9 - p_6)$
	Φ_{20}	n_{9x}^6	o_{9x}^6	a_{9x}^6	n_{9y}^6	o_{9y}^6	a_{9y}^6	n_{9z}^6	o_{9z}^6	a_{9z}^6	n_6	o_6	a_6	$(p_9 - p_6)$
Ψ_{19}	M_{19}	$n_{9x}^6 = n_6 n_9$	$o_{9x}^6 = o_6 n_9$	$a_{9x}^6 = a_6 n_9$	$n_{9y}^6 = n_6 o_9$	$o_{9y}^6 = o_6 o_9$	$a_{9y}^6 = a_6 o_9$	$n_{9z}^6 = n_6 a_9$	$o_{9z}^6 = o_6 a_9$	$a_{9z}^6 = a_6 a_9$	n_6	o_6	a_6	$(p_9 - p_6)$
	Φ_{19}	n_6	o_6	a_6	n_9	o_9	a_9	$(p_9 - p_6)$						
	Φ_{19}	n_6	o_6	a_6	n_9	o_9	a_9	p_6	p_9					
Input Parameters		n_6	o_6	a_6	n_9	o_9	a_9	p_6	p_9					

The structure and node information of the 7,9 joint prior block:

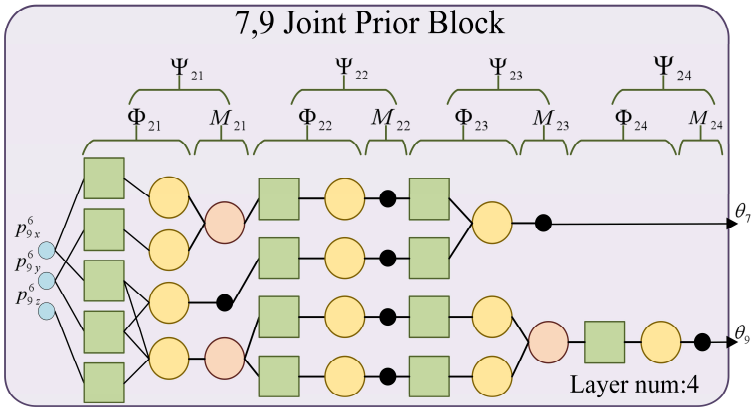


Fig. 8. The 7,9 Joint Prior Block

TABLE VIII

THE NODE INFORMATION OF 7,9 JOINT PRIOR BLOCK

Ψ_L: HMRF-KAN Layers
Φ_L: KAN Layers
M_L: Multiplication Layers

		Node Information				
Output		θ_7				θ_9
Ψ_{24}	M_{24}	θ_7				θ_9
	Φ_{24}	θ_7				θ_9
		θ_7				$\theta_9 = \text{atan2}(a_8, d_9) - \text{atan2}(f_1, \sqrt{a_8^2 + d_9^2 - f_1^2})$
Ψ_{23}	M_{23}	θ_7				$\frac{f_1}{\sqrt{a_8^2 + d_9^2 - f_1^2}}$
	Φ_{23}	$\theta_7 = \text{atan2}(p_{9y}^6, p_{9x}^6) - \text{atan2}(\text{d}_8, \sqrt{(p_{9x}^6)^2 + (p_{9y}^6)^2 - d_8^2})$		f_1		$\frac{1}{\sqrt{a_8^2 + d_9^2 - f_1^2}}$
		$\text{atan2}(p_{9y}^6, p_{9x}^6)$	$\text{atan2}(\text{d}_8, \sqrt{(p_{9x}^6)^2 + (p_{9y}^6)^2 - d_8^2})$	f_1		$\sqrt{a_8^2 + d_9^2 - f_1^2}$
Ψ_{22}	M_{22}	$\text{atan2}(p_{9y}^6, p_{9x}^6)$	$\sqrt{(p_{9x}^6)^2 + (p_{9y}^6)^2 - d_8^2}$	f_1		$\sqrt{a_8^2 + d_9^2 - f_1^2}$
	Φ_{22}	$\text{atan2}(p_{9y}^6, p_{9x}^6)$	$\sqrt{(p_{9x}^6)^2 + (p_{9y}^6)^2 - d_8^2}$	f_1		$\sqrt{a_8^2 + d_9^2 - f_1^2}$
		$\text{atan2}(p_{9y}^6, p_{9x}^6)$	$\sqrt{(p_{9x}^6)^2 + (p_{9y}^6)^2 - d_8^2}$	f_1		$\sqrt{a_8^2 + d_9^2 - f_1^2}$
Ψ_{21}	M_{21}	$p_{9\text{y}}^6 (p_{9\text{x}}^6)^{-1}$		$((p_{9x}^6)^2 + (p_{9y}^6)^2 - d_8^2)$		$f_1 = (\frac{(p_{9x}^6)^2 + (p_{9y}^6)^2 + (p_{9z}^6)^2 - a_8^2 - a_9^2 - d_8^2}{2a_8})$
	Φ_{21}	$p_{9\text{y}}^6$	$(p_{9\text{x}}^6)^{-1}$	$((p_{9x}^6)^2 + (p_{9y}^6)^2 - d_8^2)$		$((p_{9x}^6)^2 + (p_{9y}^6)^2 + (p_{9z}^6)^2 - a_8^2 - a_9^2 - d_8^2)$
		$p_{9\text{y}}^6$	$(p_{9\text{x}}^6)^{-1}$	$(p_{9x}^6)^2$	$(p_{9y}^6)^2$	$(p_{9z}^6)^2$
Input Parameters		p_{9x}^6 p_{9y}^6 p_{9z}^6				

The structure and node information of the 8 joint prior block:

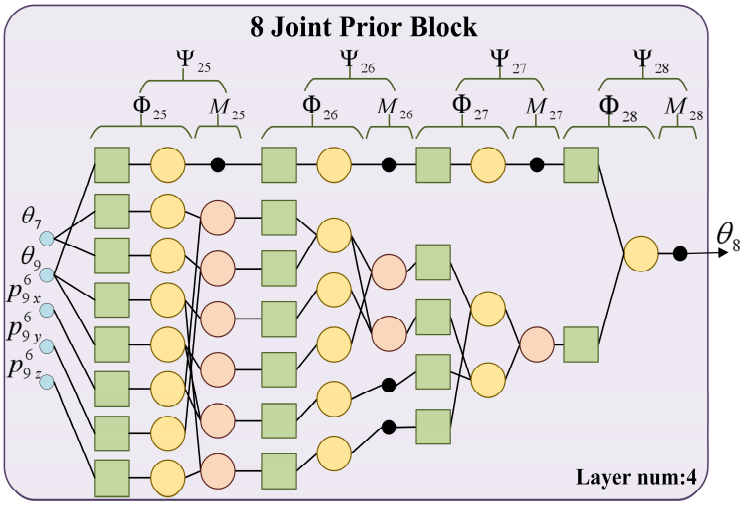


Fig. 9. The 8 Joint Prior Block

TABLE IX

THE NODE INFORMATION OF 8 JOINT PRIOR BLOCK

Ψ_L: HMRF-KAN Layers Φ_L: KAN Layers M_L: Multiplication Layers		Node Information						
Output		θ_8						
Ψ_{28}	M_{28}	θ_8						
	Φ_{28}	$\theta_8 = (\theta_8 + \theta_9) - \theta_9$						
		θ_9	$(\theta_8 + \theta_9) = \text{atan2}(f_4, f_5^{-1})$					
Ψ_{27}	M_{27}	θ_9	$f_4 \cdot f_5$					
	Φ_{27}	θ_9	$f_4 = f_2 - (a_8\cos(\theta_9) + a_9)\cdot p_{9z}^6$			$f_5 = (f_3 + (a_8\sin(\theta_9))\cdot p_{9x}^6)^{-1}$		
		θ_9	f_2		f_3	$(a_8\sin(\theta_9))\cdot p_{9x}^6$	$(a_8\cos(\theta_9) + a_9)\cdot p_{9z}^6$	
Ψ_{26}	M_{26}	θ_9	$f_2 = f_1\cdot(a_8\sin(\theta_9))$		$f_3 = f_1\cdot(a_8\cos(\theta_9) + a_9)$	$(a_8\sin(\theta_9))\cdot p_{9x}^6$	$(a_8\cos(\theta_9) + a_9)\cdot p_{9z}^6$	
	Φ_{26}	θ_9	$f_1 = \sin(\theta_7)p_{9x}^6 + \sin(\theta_7)p_{9y}^6$	$a_8\cos(\theta_9) + a_9$	$a_8\sin(\theta_9)$	$(a_8\sin(\theta_9))\cdot p_{9x}^6$	$(a_8\cos(\theta_9) + a_9)\cdot p_{9z}^6$	
		θ_9	$\sin(\theta_7)p_{9x}^6$	$\sin(\theta_7)p_{9y}^6$	$a_8\cos(\theta_9) + a_9$	$a_8\sin(\theta_9)$	$(a_8\sin(\theta_9))\cdot p_{9x}^6$	$(a_8\cos(\theta_9) + a_9)\cdot p_{9z}^6$
Ψ_{25}	M_{25}	θ_9	$\sin(\theta_7)p_{9x}^6$	$\sin(\theta_7)p_{9y}^6$	$a_8\cos(\theta_9) + a_9$	$a_8\sin(\theta_9)$	$(a_8\sin(\theta_9))\cdot p_{9x}^6$	$(a_8\cos(\theta_9) + a_9)\cdot p_{9z}^6$
	Φ_{25}	θ_9	$\cos(\theta_7)$	$\sin(\theta_7)$	$a_8\cos(\theta_9) + a_9$	$a_8\sin(\theta_9)$	p_{9x}^6	p_{9y}^6 p_{9z}^6
		θ_9	$\cos(\theta_7)$	$\sin(\theta_7)$	$a_8\cos(\theta_9)$	$a_8\sin(\theta_9)$	p_{9x}^6	p_{9y}^6 p_{9z}^6
Input Parameters		θ_7 θ_9 p_{9x}^6 p_{9y}^6 p_{9z}^6						

S2：Detailed discussion on the modified DH parameters of 9-DOF manipulator structure in *II-E*

Fig. 10 shows the structure of the 9-DOF manipulator, and Table X lists the corresponding modified DH parameters.

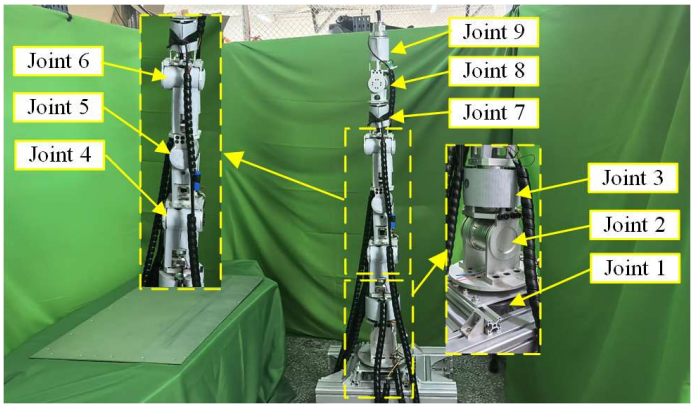


Fig. 10. The Structure of the 9-DOF Manipulator

TABLE X

THE MODIFIED DH PARAMETERS OF 9-DOF MANIPULATOR

Joint	α_i (rad)	a_i (mm)	θ_i (rad)	d_i (mm)
1	0	0	0	496
2	$\frac{\pi}{2}$	0	0	3.5
3	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	312
4	$-\frac{\pi}{2}$	0	$-\frac{\pi}{2}$	8.5
5	$-\frac{\pi}{2}$	201	0	9.5
6	$-\frac{\pi}{2}$	196	$\frac{\pi}{2}$	8.5
7	$\frac{\pi}{2}$	0	$\frac{\pi}{2}$	214.5
8	$\frac{\pi}{2}$	0	0	1
9	$-\frac{\pi}{2}$	0	0	151.5